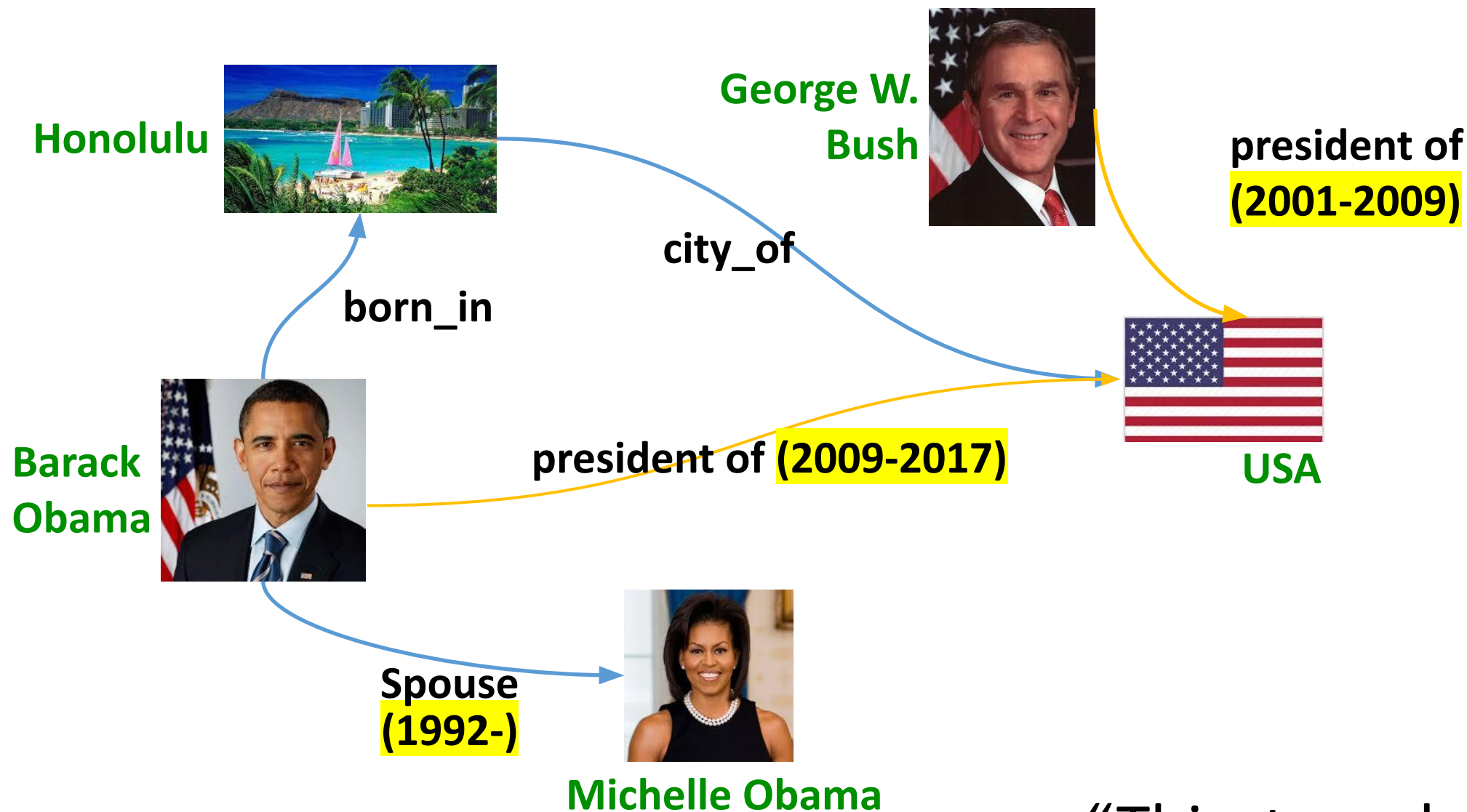


Knowledge and Retrieval

Temporal Knowledge Graphs

Temporal Knowledge Graphs



“This, too, shall pass”₂

Temporal KG completion: tasks

- Temporal tuple
(subject, relation, object, begin, end)
 - E.g. (Obama, president-of, USA, 2009, 2017)

We limit to single intervals or recurrent instants

- **Link prediction task**

(?, rel, obj, begin, end) or (subj, rel, ?, begin, end)

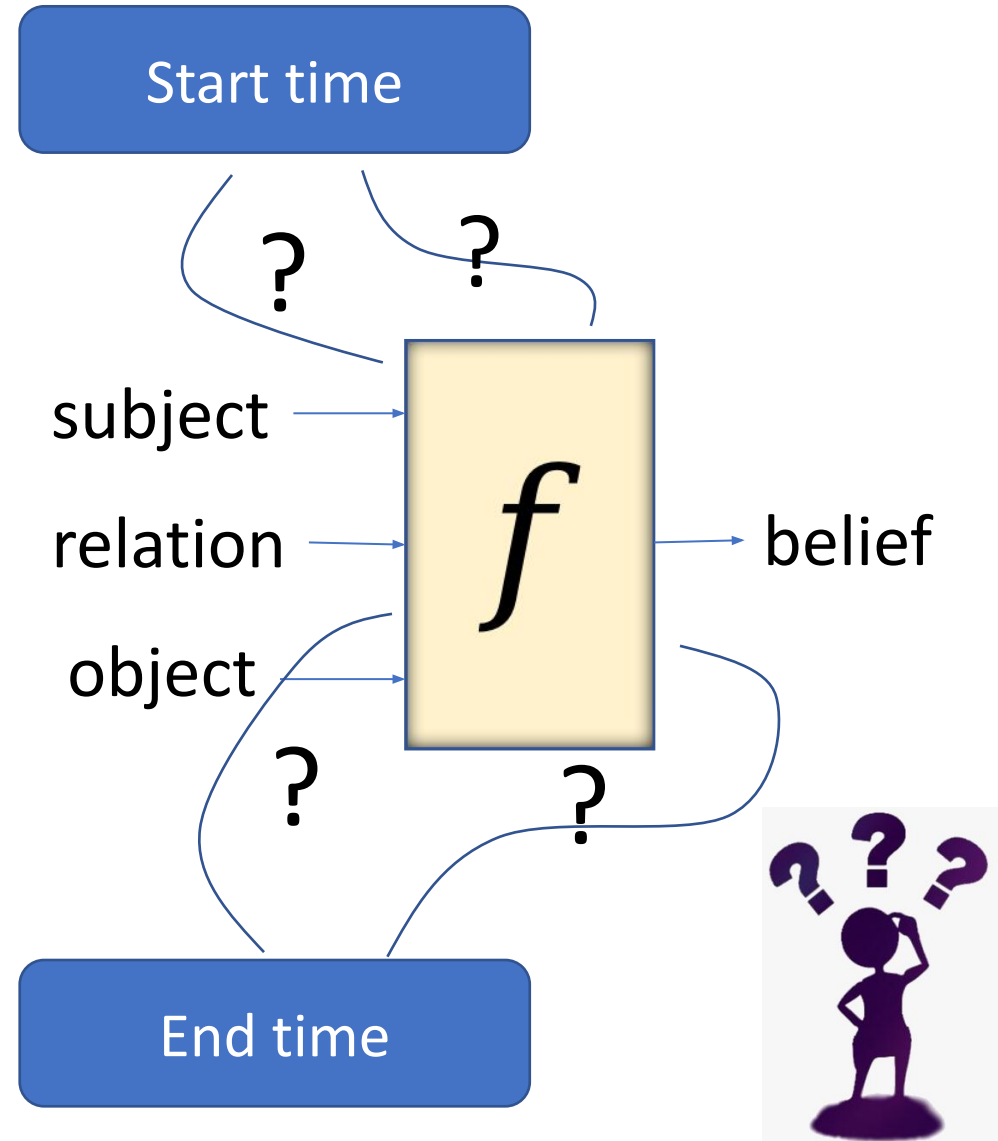
- (Barak Obama, president of , *USA*, 2009, 2017) ☐ positive
- (Barak Obama, president of , *India*, 2009, 2017) ☐ negative
- (*Barak Obama*, president of , USA, 2009, 2017)
- (*George W. Bush*, president of , USA, 2009, 2017)

- **Time prediction task** (subject, relation, object, ?, ?)

- (Barak Obama, president of , USA, *2009*, *2017*)
- (Barak Obama, president of , USA, *2001*, *2009*)

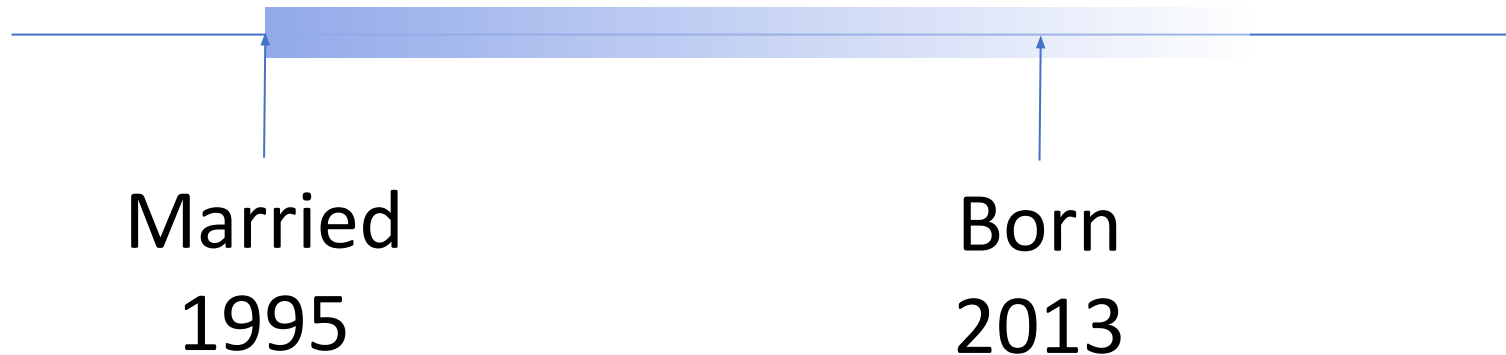
Extend KGC to temporal KGC?

- Earlier s , r , o were arguments to score function f
- Now start and end times are additional arguments
- Where do these fit in?
- How should f be redesigned or extended?
- Will entities and relations get time-varying representations?



Time prediction: Failure sample

- In train fold:
Shin_Ae-ra, isMarriedTo, ChaIn-Pyo, (1995, -)⟩
- Test fold query:
Shin_Ae-ra, bornInYear, ____⟩
- A recent temporal KGC method responds with...



HyTE: time as hyperplanes

- Entity and relation embeddings e_h , e_r , e_t live in a space-time continuum
- Each time instant τ is associated with a subspace
 - Specifically, a hyperplane
- Entity and relation embeddings projected to time hyperplane, as $e_h(\tau)$, $e_r(\tau)$, $e_t(\tau)$
- In the hyperplane, TransE model applies: $e_h(\tau) + e_r(\tau) \approx e_t(\tau)$
- Negative sampling extended to time

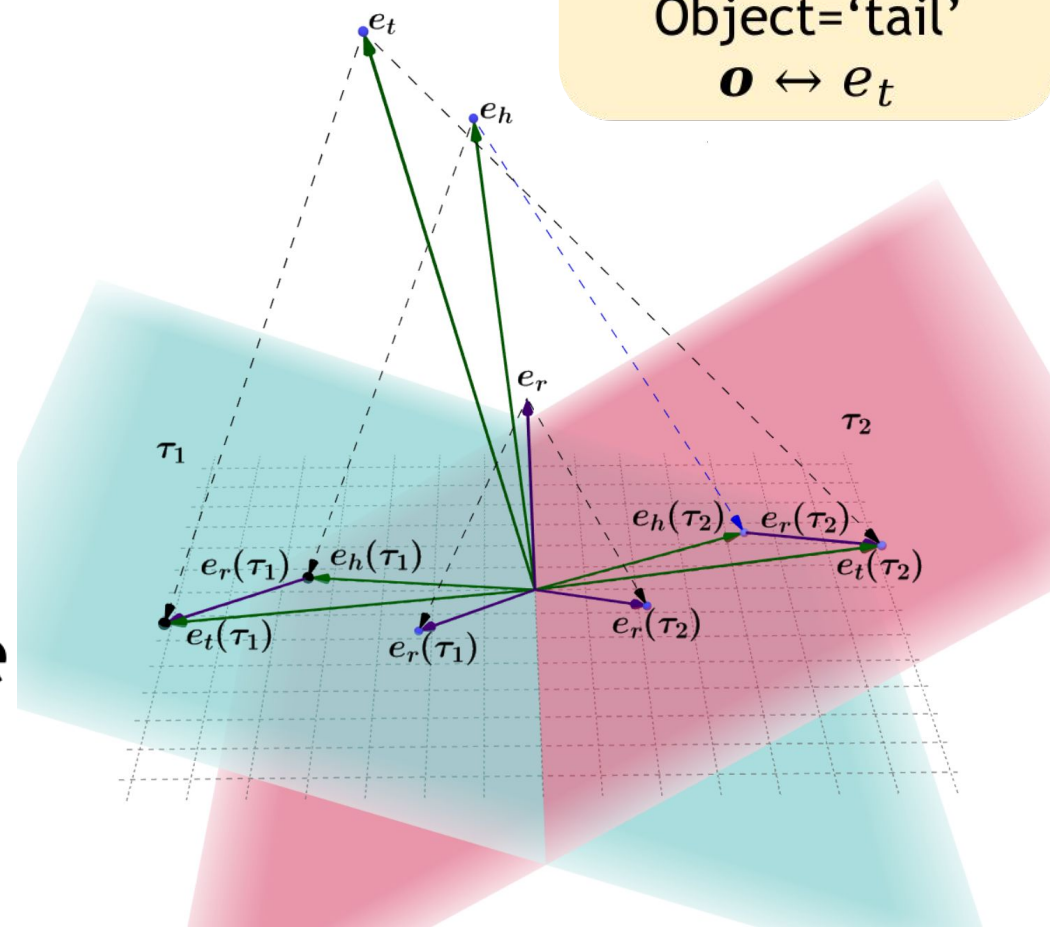
Subject='head'

$\mathbf{s} \leftrightarrow e_h$

Relation $\mathbf{r} \leftrightarrow e_r$

Object='tail'

$\mathbf{o} \leftrightarrow e_t$

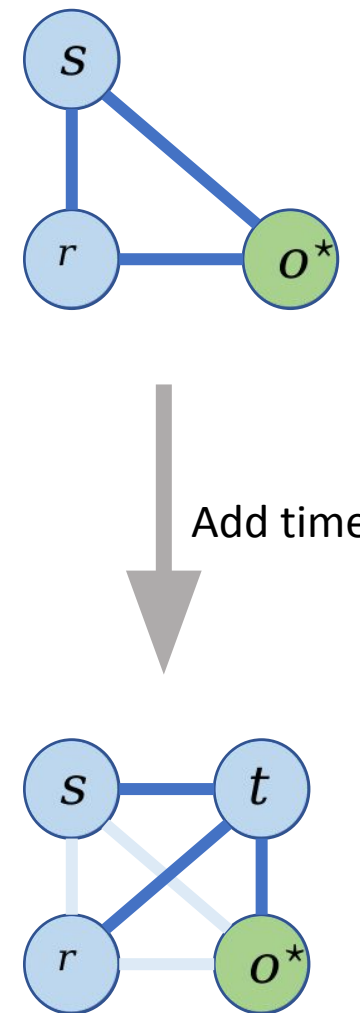


As in TransH

Prefer stronger baseline

Extending ComplEx with time

- Recall ComplEx score $\Re\langle \mathbf{s}, \mathbf{r}, \mathbf{o}^* \rangle$
- Embed time t to $\mathbf{t} \in \mathbb{C}^D$
- TComplEx score $\Re\langle \mathbf{s}, \mathbf{r}, \mathbf{o}^*, \mathbf{t} \rangle$
 - Equivalent to any of $\Re\langle \mathbf{s} \odot \mathbf{t}, \mathbf{r}, \mathbf{o}^* \rangle$, $\Re\langle \mathbf{s}, \mathbf{r} \odot \mathbf{t}, \mathbf{o}^* \rangle$, and $\Re\langle \mathbf{s}, \mathbf{r}, \mathbf{o}^* \odot \mathbf{t} \rangle$
 - $\odot$ is elementwise multiplication
 - Time modulates exactly one of subject, relation, object



Mixing ephemeral and eternal

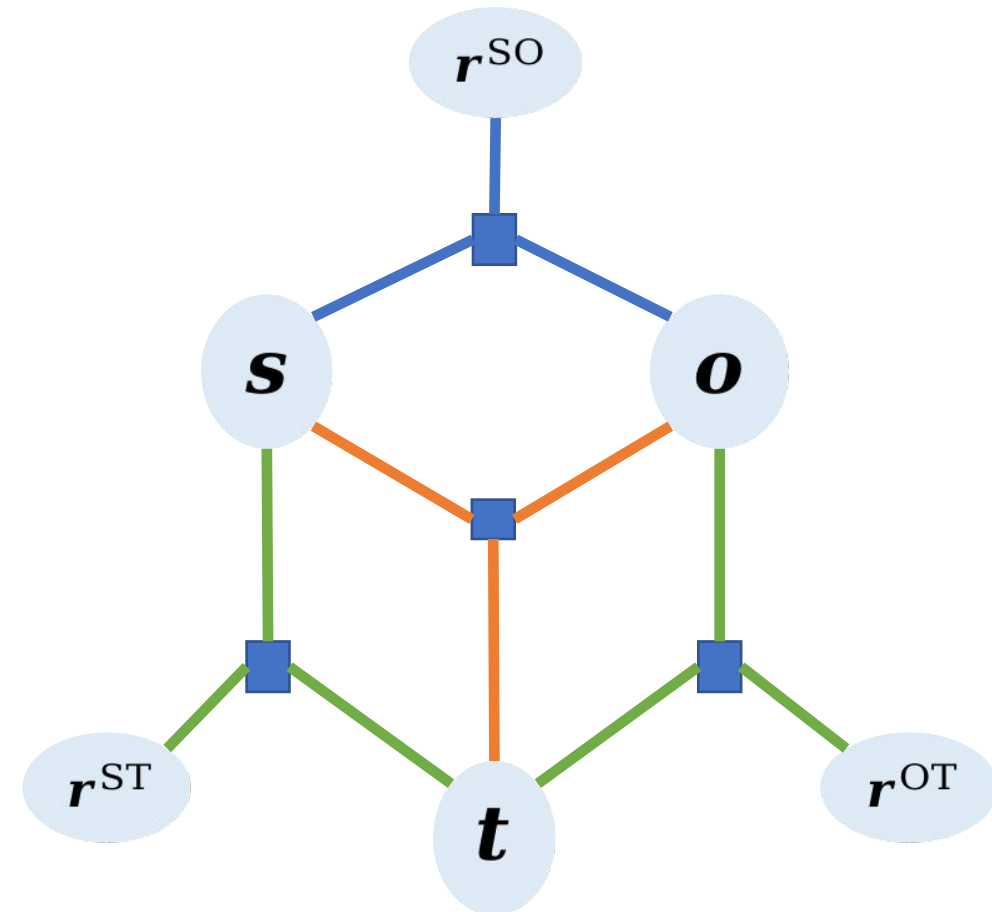
- TNTComplEx score mixes time-sensitive and time-independent parts

$$\Re(\langle \mathbf{s}, \mathbf{r}_T, \mathbf{o}^*, \mathbf{t} \rangle + \langle \mathbf{s}, \mathbf{r}, \mathbf{o}^* \rangle)$$

- $\mathbf{r}_T$ is time-sensitive relation embedding
- $\mathbf{r}$ is time-independent relation embedding
- Equivalent to $\Re(\langle \mathbf{s}, \mathbf{r}_T \odot \mathbf{t} + \mathbf{r}, \mathbf{o}^* \rangle)$
- Training reveals what kind of relation r is
- TNTComplEx generally better than HyTE

Timeline factors

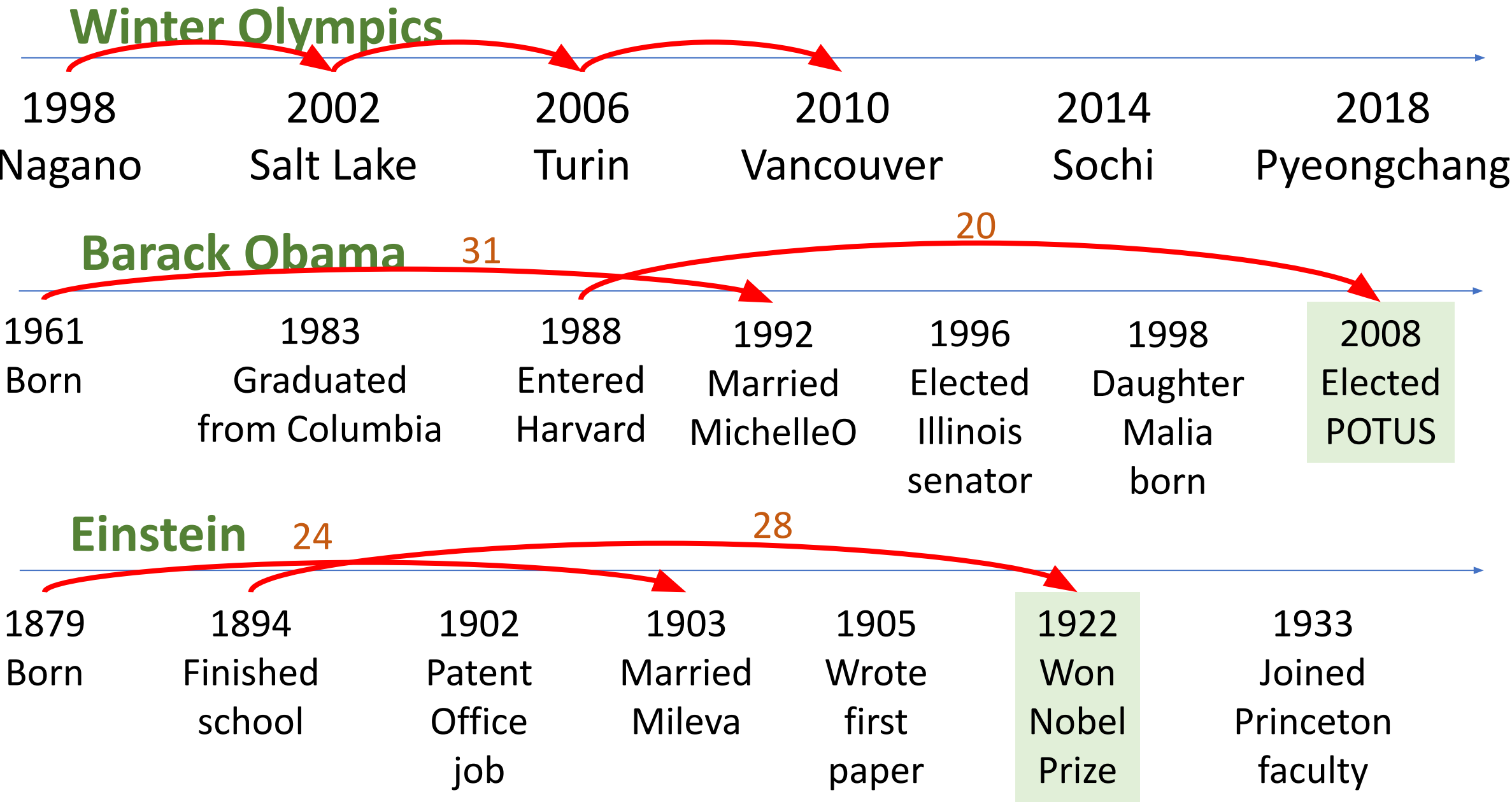
- r^{SO} , which was r earlier, scores if s and o can be related by r , independent of time
- Can s participate in relation r at time t ? $\rightarrow r^{ST}$
 - President of * at age 10
- Can o participate in relation r at time t ? $\rightarrow r^{OT}$
 - * of Google in 1980
- Can s and o interact via any relation at time t ?
 - Mozart, Elvis, 2005



TimePlex (base)

- Reuse 3-factor score function from ComplEx
- $\Re(\langle \mathbf{s}, \mathbf{r}^{\text{SO}}, \mathbf{o}^* \rangle + \blacksquare \langle \mathbf{s}, \mathbf{r}^{\text{ST}}, \mathbf{t}^* \rangle + \blacksquare \langle \mathbf{o}, \mathbf{r}^{\text{OT}}, \mathbf{t}^* \rangle + \blacksquare \langle \mathbf{s}, \mathbf{o}, \mathbf{t}^* \rangle)$
 - Balancing weight hyperparameters
- This base model is better than TComplEx and TNTComplEx
- Two more signals still to exploit...

History may not repeat, but it rhymes



Temporal constraints

• Fact recurrence:

- Olympic games, elections, league matches

Non-recurrence:

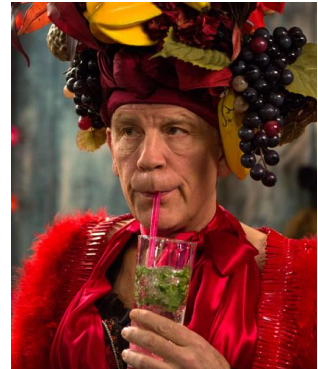
- A person is born, dies only once

Ordering between relations:

- Must be born before marrying, dying

Distribution over time gaps between relations:

- $e.\text{diedIn}.T - e.\text{wasBornIn}.T \approx 70 \text{ years}$



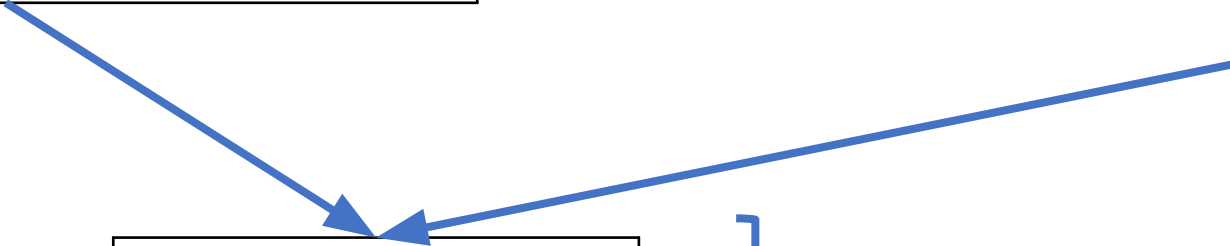
Collecting time gap statistics

r : bornIn

Joe Biden, bornIn, USA, 1942
Michelle Obama, bornIn, USA, 1964
A.P.J. Abdul Kalam, bornIn, India, 1931

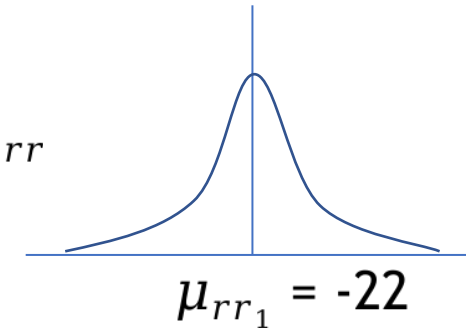
r_1 : graduatedFrom

Joe Biden, graduatedFrom, University of Delaware, 1965
Michelle Obama, graduatedFrom, Princeton University, 1985
A.P.J. Abdul Kalam, graduatedFrom, University of Madras, 1954



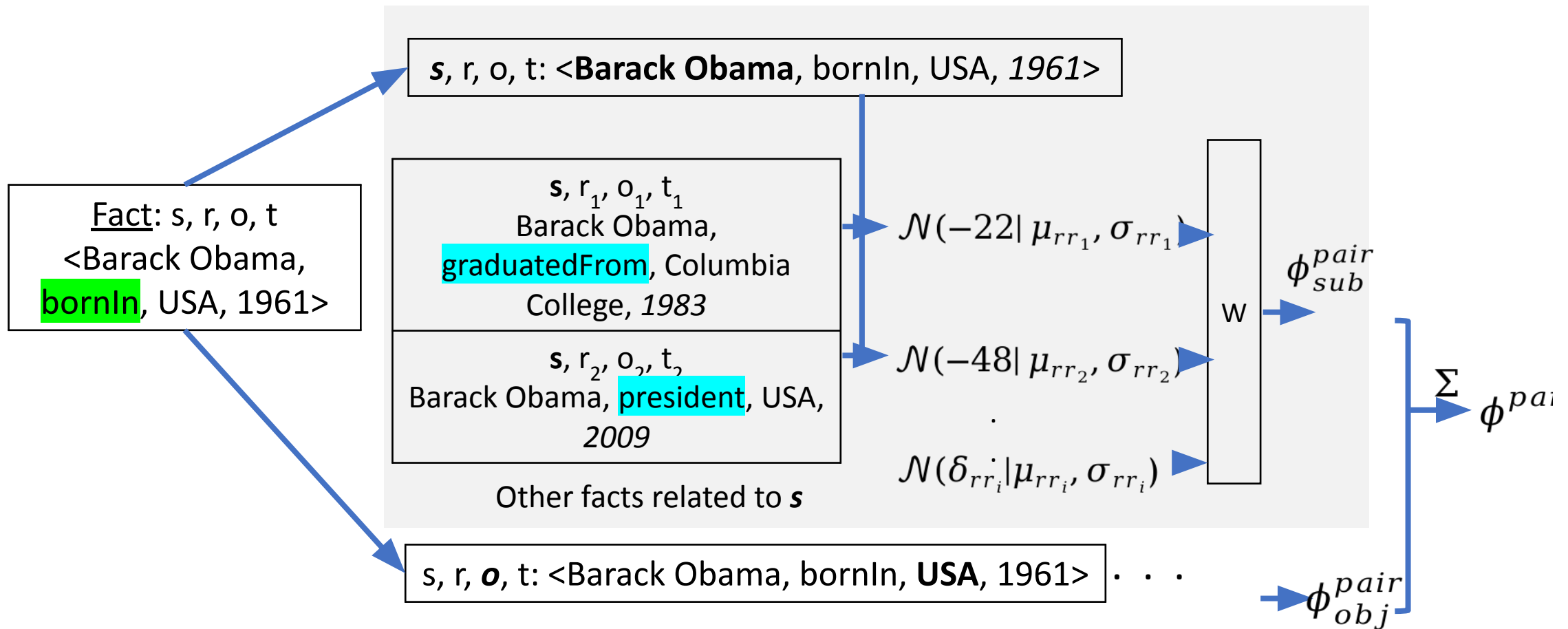
$(1942 - 1965) = -23$
$(1964 - 1985) = -21$
$(1931 - 1954) = -23$

$\mathcal{N}(\mu_{rr_1}, \sigma_{rr})$



Distribution of number of years from birth to college graduation

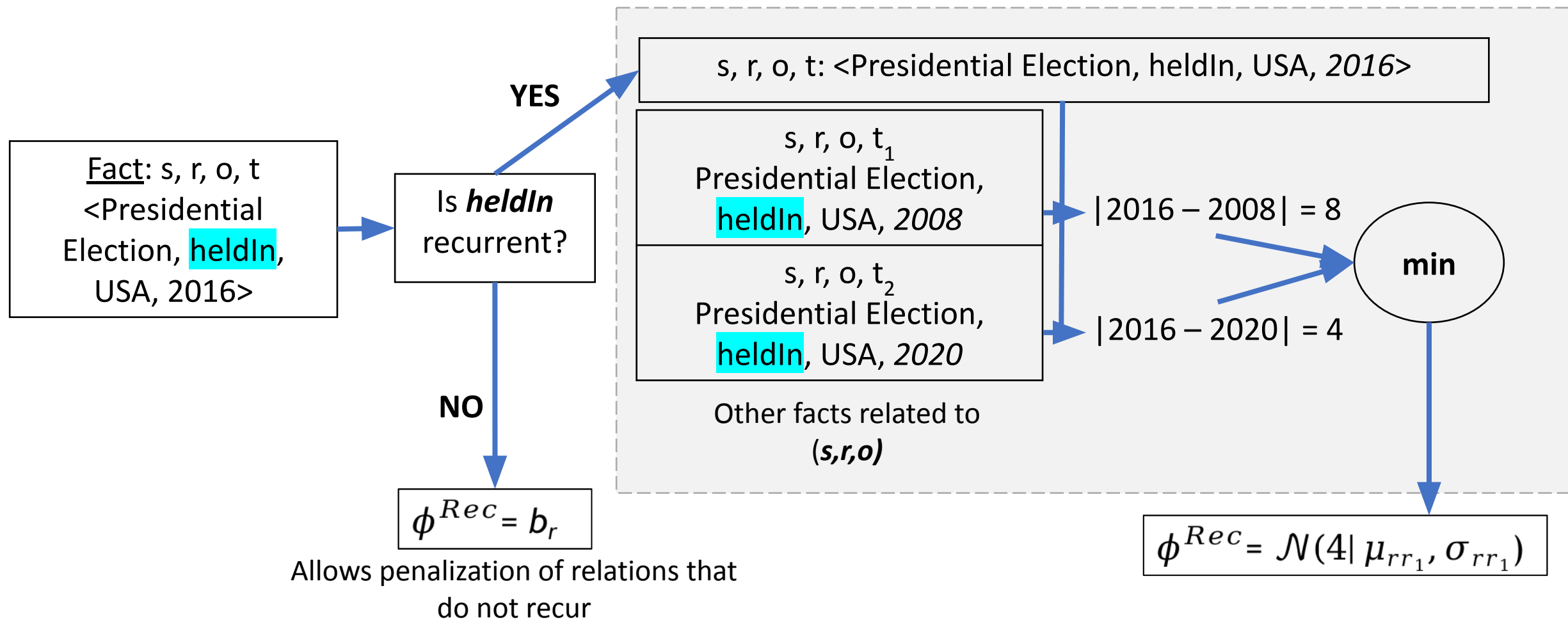
Modeling relation gap constraints



How likely is it that Obama was born in 1961?

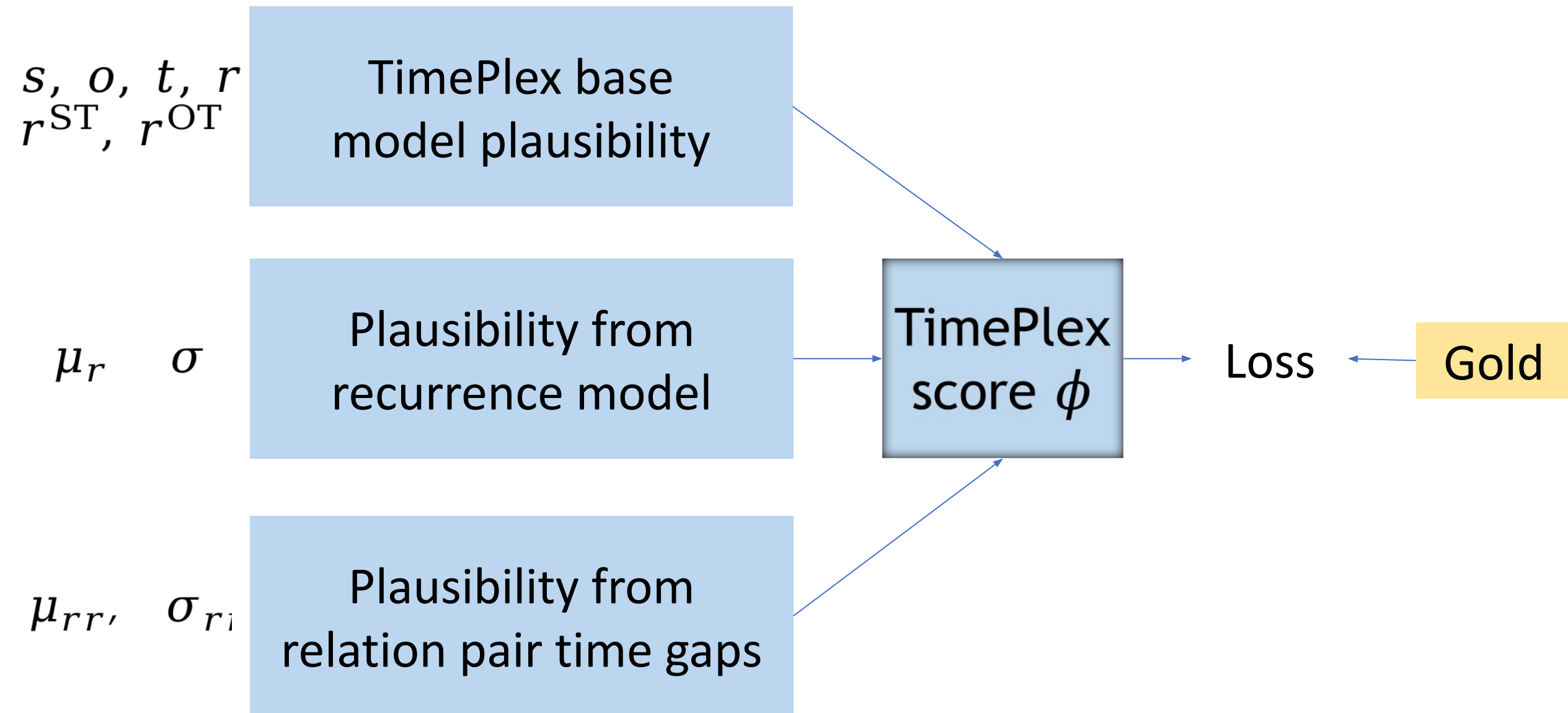
Depends on when he graduated from college, when he became president, ...

Modeling recurrence



Was a presidential election held in USA in 2016?
Depends ... on nearby years when it was held

Combining the signals



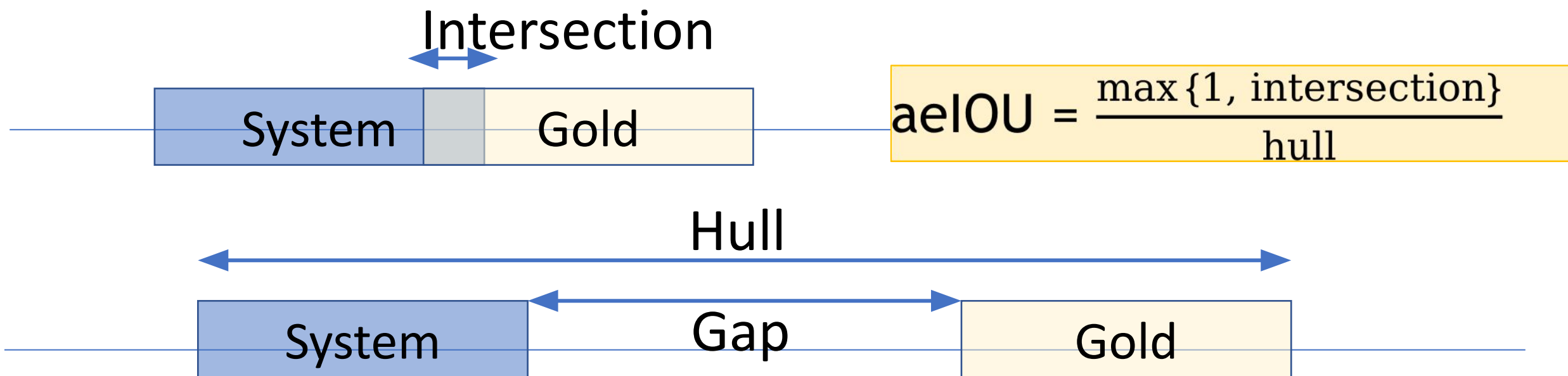
Link Prediction Performance

Dataset→	WIKIDATA12k			YAGO11k			ICEWS05-15			ICEWS14		
↓Methods	MRR	HITS@1	HITS@10	MRR	HITS@1	HITS@10	MRR	HITS@1	HITS@10	MRR	HITS@1	HITS@10
CX	24.82	14.30	48.90	18.14	11.46	31.11	48.68	37.00	72.63	45.50	33.87	69.73
TNT-Complex	30.10	19.73	50.69	18.01	11.02	31.28	60.58	51.14	78.50	56.72	47.04	75.40
TIMEPLEX (base)	32.38	22.03	52.79	18.35	10.99	31.86	63.91	54.62	81.42	60.25	51.29	77.05
TIMEPLEX	33.35	22.78	53.20	23.64	16.92	36.71	63.99	54.51	81.81	60.40	51.50	77.11

- TimePlex variants best on four diverse TKBC benchmarks
- Evaluation measure design is nontrivial
 - What if train fold revealed a fraction of valid interval?

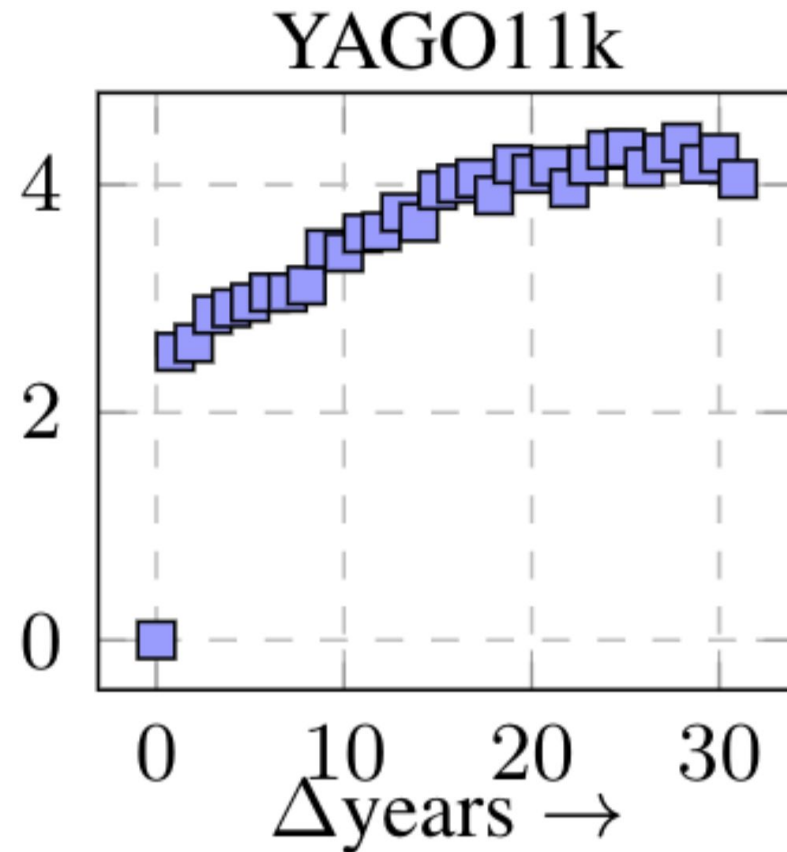
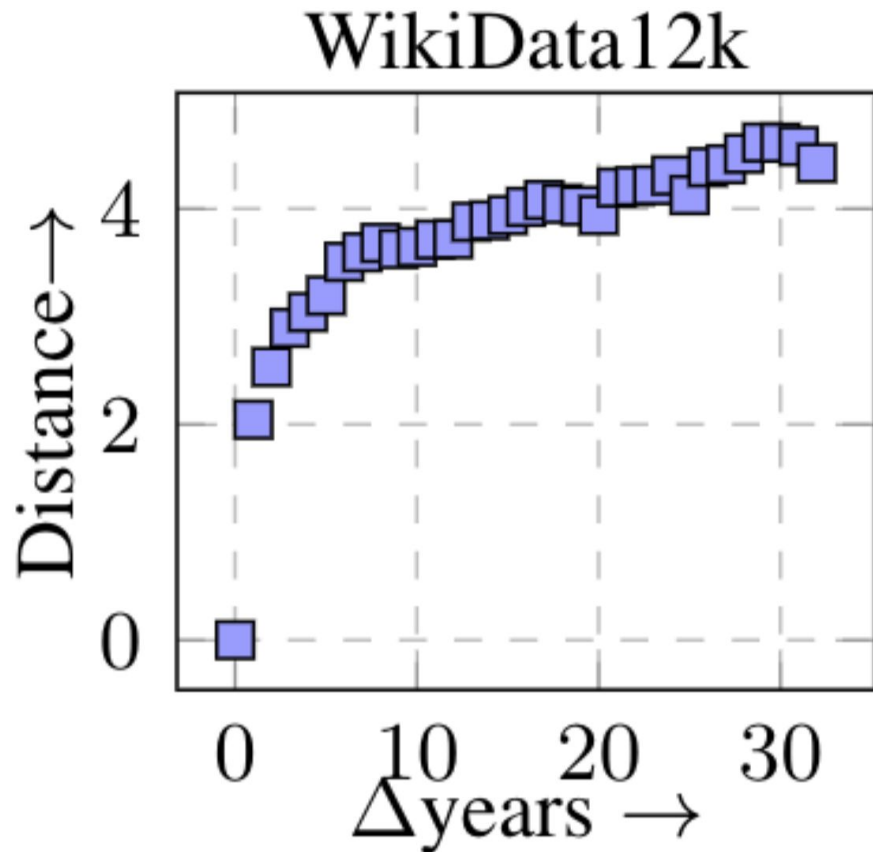
Time Interval Prediction Performance

Datasets→	YAGO11k	WIKIDATA12k
↓Methods	aeIOU	aeIOU
TNT-Complex	8.40	23.35
TIMEPLEX (base)	14.21	26.20
TIMEPLEX	20.03	26.36

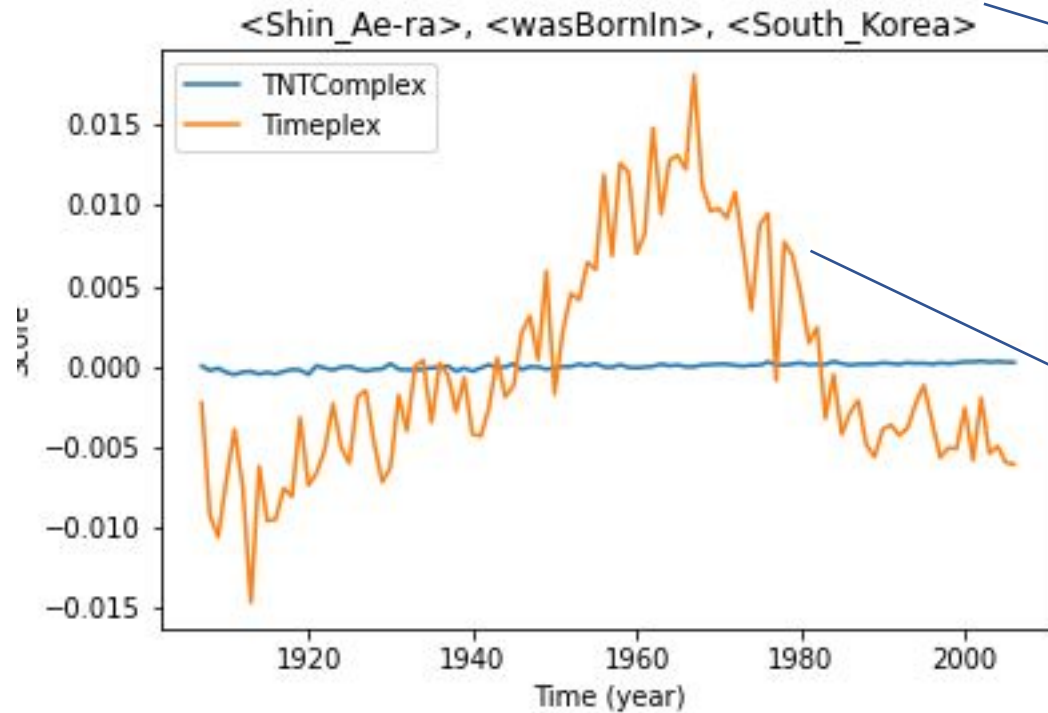


Time gap vs. embedding distances

- Fix a year; compare its embedding to that of other years
- Distance increases with increasing year difference
- No explicit attempt to train a loss to this end!



Time prediction profile



Query: when was a person
born (gold=1969)

TimePlex belief along time
(TNT-Complex is flat)

Train KB:

TimePlex prediction: 1967; aelOU ~ 0.33

TNT-Complex prediction: 2013; aelOU ~ 0

<Shin_Ae-ra, isMarriedTo, ChaIn-Pyo, (1995, -)>

Known: married in 1995

Summary

- Feverish pace of recent work on temporal KGs
- We just scratched the surface
- No matter how much training data we throw ...
- ... at a generic network for score f ,
- it seems unlikely that commonsense knowledge about durations, time gaps, and recurrences will surface automatically in deep models ...
- ... without a little help from outside